DIP Lab Assignment 03

Objective

The objective of this experiment is to understand and implement histogram equalization using the Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of a grayscale image.

Task

You are given a digital image of your choice.
Perform the following operations on the image:

Step 1: Read the Image
convert it into a grayscale image.
Do not use a completely uniform or completely black/white image.

Step 2: Calculate the Histogram
For the grayscale image, calculate the frequency of each intensity level.
Plot the histogram of the grayscale image.

Step 3: Calculate the PDF
Calculate the Probability Density Function from the histogram.
Display or print the PDF values.

Step 4: Calculate the CDF
Calculate the Cumulative Distribution Function from the PDF.
Plot the CDF.

Step 5: Perform Histogram Equalization
Use the CDF to calculate the new intensity value for every pixel.
Use the resulting mapping to create the histogram-equalized image.

Step 6: Calculate the Equalized Histogram
Calculate the histogram of the equalized image.

Display:

1. Original grayscale image
2. Histogram of original image
3. PDF of original image
4. CDF of original image
5. Histogram-equalized image
6. Histogram of equalized image

Output	Requirement
Original Image	Display
Grayscale Image	Display
Original Histogram	Plot
PDF	Calculate and display
CDF	Calculate and plot
Equalized Image	Display
Equalized Histogram	Plot

You must implement the PDF, CDF, and histogram equalization calculations yourselves.

Do not directly use a built-in function such as:

- equalizeHist()
- or any other function that directly performs histogram equalization.

You may use libraries for:

- Reading images
- Converting RGB images to grayscale
- Displaying images
- Plotting graphs
- Basic numerical operations

Submit your complete source code.